MARRI LAXMAN REDDY
INSTITUTE OF TECHNOLOGY AND MANAGEMENT
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AI-Enabled Smart Battery Management System for Electric Vehicles with State-of-Health Prediction, Fault Detection, and Safe Adaptive Charging.

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INTRODUCTION

- EV batteries are the most critical and expensive part of an electric vehicle.
- Battery safety, health, and charging efficiency are major concerns.
- Conventional BMS uses fixed thresholds for voltage, current, and temperature.
- AI can help predict battery health and detect faults earlier.
- This project combines sensing, AI prediction, and safe charging control.

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PROBLEM STATEMENT

- Conventional BMS may detect faults only after limits are crossed.
- Weak cells and gradual degradation may not be identified early.
- Fixed charging methods do not adapt to battery health and temperature.
- A low-cost hardware prototype with AI-based decision making is needed.

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OBJECTIVES

- To monitor battery voltage, current, and temperature in real time.
- To predict State of Health using machine learning.
- To detect battery faults such as overtemperature, overcurrent, and cell imbalance.
- To implement safe adaptive charging control.
- To provide a dashboard for monitoring and alerts.
- To validate the prototype using a low-voltage battery setup.

EXISTING SYSTEM

- Fixed threshold-based battery protection.
- Pack-level monitoring only.
- No early prediction of battery degradation.
- No intelligent charging decisions.
- Limited adaptability under changing battery conditions.

Limitations:

- Poor early warning.
- No confidence-based decision making.
- No cell-level health analysis

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LITERATURE SURVEY

PAPER	MAIN IDEA	LIMITATIONS
Intelligent Battery Management System with AI and IoT for Enhanced EV Efficiency	AI + IoT based monitoring and predictive maintenance	Mostly system-level, limited prototype depth
Hybrid ML Model for EV Battery SoC and SoH Prediction	Predicts SoC and SoH	Focuses mainly on estimation
Deep Learning-Based Battery Health Prediction	High-accuracy battery health prediction	Computationally complex
AI/ML-Enabled Smart Battery Management Systems in EVs	Review of AI methods in BMS	Identifies gaps, does not build prototype

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PROPOSED SYSTEM

- Low-voltage prototype using ESP32 and battery sensors.
- Cell-level voltage monitoring.
- Random Forest model for SoH prediction.
- Random Forest model for fault detection.
- Safe adaptive charging using predefined modes.
- Hardware protection remains active independently.
- AI confidence can be used to switch to safe fallback mode.

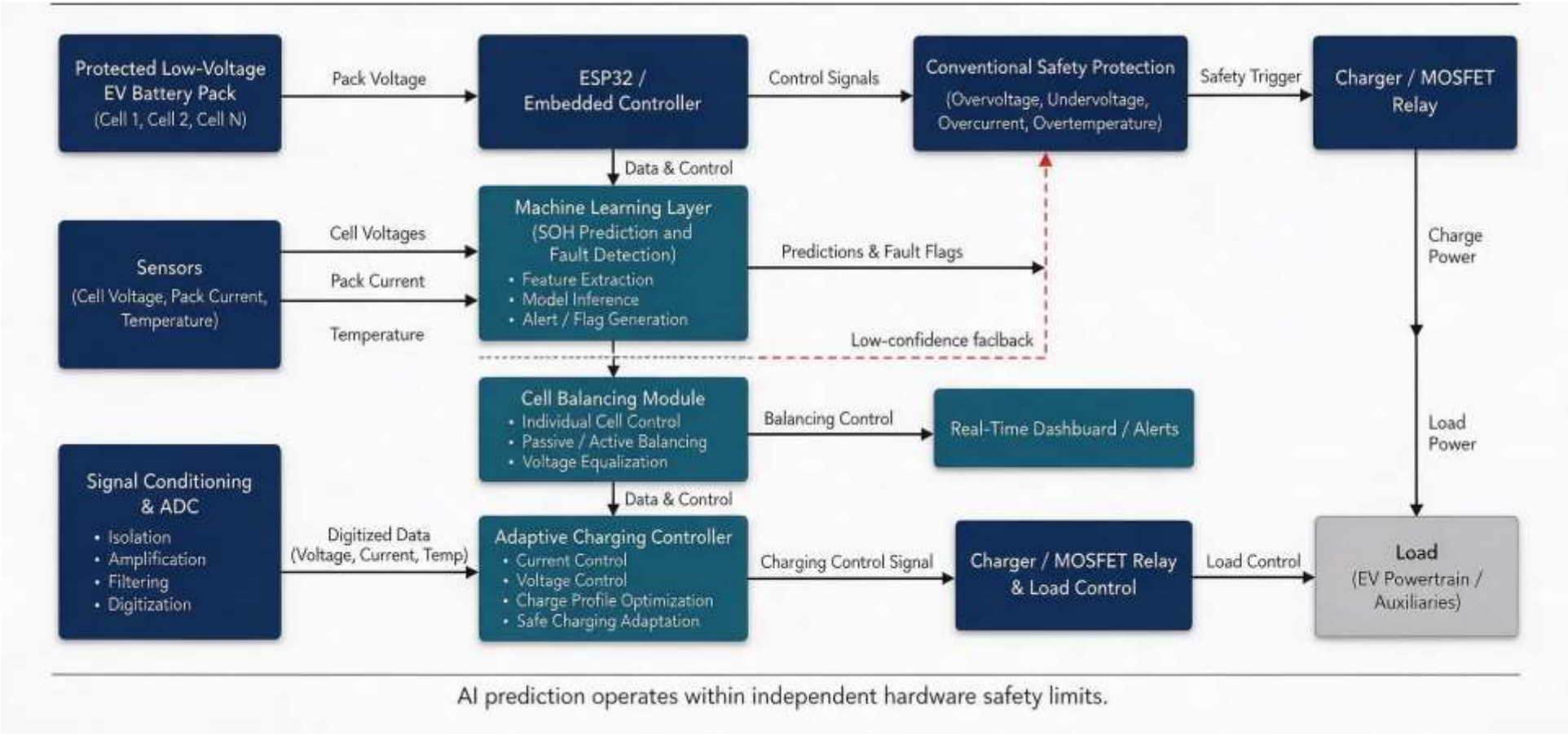
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METHODOLOGY

1. Battery voltage, current, and temperature are measured.
2. Sensor data is stored and pre-processed.
3. Dataset is used to train the AI model.
4. SoH and fault classes are predicted.
5. Charging mode is selected based on prediction.
6. Safety logic checks hardware limits.
7. Charger is controlled through relay/MOSFET.
8. Results are shown on OLED/dashboard

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BLOCK DIAGRAM



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EXISTING SYSTEM

- Measures battery voltage, current, and temperature.
- Estimates State of Charge using voltage or current-based methods.
- Protects the battery from:
  - Overcharging.
  - Overcurrent.
  - Overtemperature.
- Performs passive cell balancing.
- Disconnects the battery when a parameter crosses a predefined limit.

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PROPOSED SYSTEM

- Measure individual cell voltage, pack current, and temperature.
- Send sensor data to an ESP32 or similar microcontroller.
- Estimate battery State of Health using an ML model.
- Detect abnormal conditions and possible faults.
- Identify cell imbalance and weak-cell behaviour.
- Select a safe charging mode based on battery condition.
- Display real-time data on a dashboard.
- Use hardware cut-off protection as a backup to the AI system.

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AI USAGE

- Cell voltage.
- Pack voltage.
- Charging and discharging current.
- Battery temperature.
- Charging time.
- Capacity delivered.
- Cycle number.
- Voltage difference between cells.

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ADAPTIVE CHARGING

Battery condition	Proposed charging response
Healthy battery and normal temperature	Normal charging mode.
Moderate cell imbalance	Reduced charging current and balancing enabled.
High temperature	Charging current reduced or charging paused.
Low predicted SOH	Conservative charging mode.
Detected fault	Charger disconnected and warning generated.
Low AI confidence or sensor failure	AI decision disabled and hardware safety rules used.

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EXPECTED OUTCOMES

- Real-time battery monitoring.
- SoH prediction with acceptable accuracy.
- Early detection of battery faults.
- Safe charging control.
- Reduced risk of overheating and imbalance.
- User-friendly dashboard display.
- Comparison with conventional BMS.

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CHALLANGES

- Limited battery data for training.
- Prototype is low-voltage, not a real EV pack.
- AI may need retraining for different battery types.
- Real-time deployment on microcontroller may be limited.
- Safety testing must be done carefully.

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FUTURE SCOPE

- Add more cell-level balancing features.
- Use a stronger dataset for training.
- Deploy Edge-AI on ESP32/Raspberry Pi.
- Add CAN communication.
- Add remaining useful life prediction.
- Expand to EV motor/load simulation.
- Improve dashboard and mobile alerts

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REFERENCES

- Intelligent Battery Management System with AI and IoT for Enhanced EV Efficiency.
- Hybrid Machine Learning Model for EV Battery SoC and SoH Prediction.
- Deep Learning-Based Battery Health Prediction for Enhancing EV Performance.
- AI/ML-Enabled Smart Battery Management Systems in EVs.
- Role of Deep Learning in Battery Management System for EVs.

THANK YOU

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